Computational Physics 2 Classes No.6 - Report

Wave equation for a 2D membrane (FEM version)

Natalia Kowalczyk

May 19, 2026

1 Introduction

In this project the classical wave equation for a two-dimensional square membrane was solved numerically using the finite element method (FEM). The membrane occupies the region $(x, y) \in (0, L) \times (0, L)$ with $L = 5$, while the wave propagation velocity was set to $v = 1$. The system is described by the equation

$$\frac{\partial^2 \Psi}{\partial t^2} = v^2 \nabla^2 \Psi,$$

where $\Psi(x, y, t)$ denotes the displacement of the membrane from its equilibrium position.

The spatial discretization was performed with bilinear finite elements. The membrane displacement was expanded in the basis of shape functions,

$$\Psi(x, y, t) = \sum_{i=1}^{N} c_i(t) h_i(x, y),$$

which transformed the continuous problem into a system of coupled ordinary differential equations for the coefficients $c_i(t)$. Time evolution was calculated using an explicit central-difference scheme.

Dirichlet boundary conditions were imposed at the outer edges of the membrane. Additionally, the central node of the mesh was forced periodically according to

$$c_{\text{center}}(t) = \sin(\omega t),$$

which allowed the study of forced oscillations and resonance effects for different driving frequencies.

The purpose of this exercise was to determine the stability condition for the time step Δt, visualize membrane vibrations, and analyze the long-time behavior of the system for several excitation frequencies.

2 Results

The calculations were performed for a square membrane of size $L = 5$ using bilinear finite elements on a mesh with $N = 10$. The wave propagation velocity was set to $v = 1$. The time evolution of the system was calculated using the explicit central-difference scheme

$$\mathbf{O}\mathbf{c}^{n+1} = \left(2\mathbf{O} - v^2 \Delta t^2 \mathbf{S}\right) \mathbf{c}^n - \mathbf{O}\mathbf{c}^{n-1},$$

where $\mathbf{O}$ denotes the overlap matrix and $\mathbf{S}$ is the stiffness matrix obtained from the FEM discretization of the Laplace operator.

The membrane boundaries were fixed, corresponding to Dirichlet boundary conditions,

$$\Psi(x, y, t) = 0$$

at the edges of the computational domain. Additionally, the central node of the mesh was forced periodically according to

$$c_{\text{center}}(t) = \sin(\omega t).$$

The stability condition for the explicit time integration method was determined numerically from the largest eigenvalue of the generalized eigenproblem

$$\mathbf{Su} = \lambda \mathbf{Ou}.$$

A stable time step satisfying the Courant-type condition was used throughout all simulations.

2.1 Oscillations for $\omega = \pi/L$

The first investigated driving frequency was

$$\omega = \frac{\pi}{L}.$$

Figure 1 presents the membrane displacement represented as a heatmap, while Fig. 2 shows the corresponding three-dimensional surface plot. A standing-wave structure develops due to reflections from the fixed boundaries and interference between different oscillation modes.

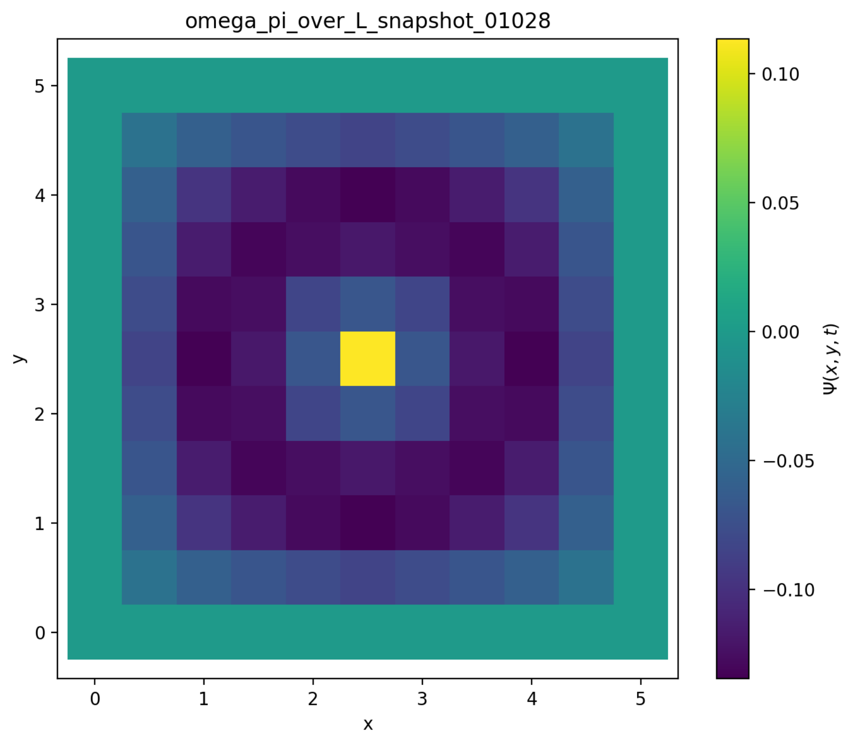

Figure 1: Heatmap of the membrane displacement for $\omega = \pi/L$.

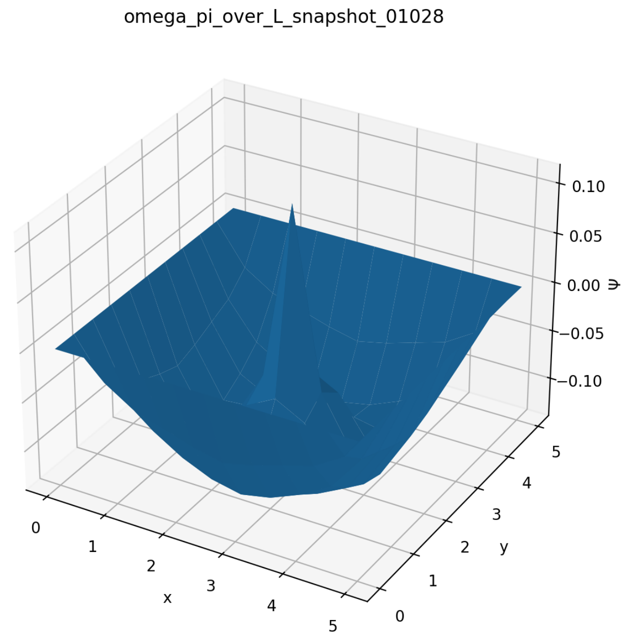

Figure 2: Three-dimensional surface plot of the membrane displacement for $\omega = \pi/L$.

The evolution of the maximal membrane amplitude is shown in Fig. 3. Since no damping term is present in the equation of motion, the oscillation amplitude grows continuously due to the external forcing.

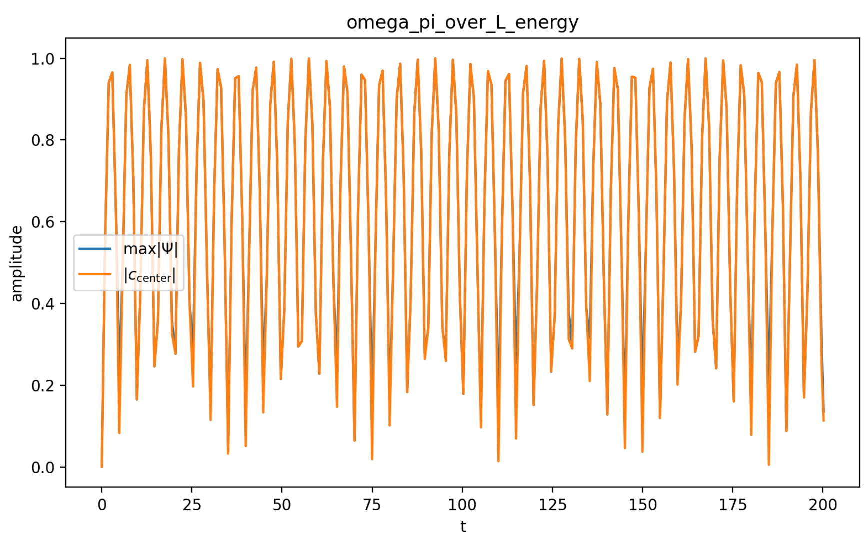

Figure 3: Maximum membrane amplitude as a function of time for $\omega = \pi/L$.

2.2 Oscillations for $\omega = 2\pi/L$

The second simulation was performed for the higher forcing frequency

$$\omega = \frac{2\pi}{L}.$$

The resulting oscillation pattern exhibits stronger spatial variations and larger amplitudes. This behavior indicates that the forcing frequency is closer to one of the membrane eigenfrequencies, leading to stronger resonance effects.

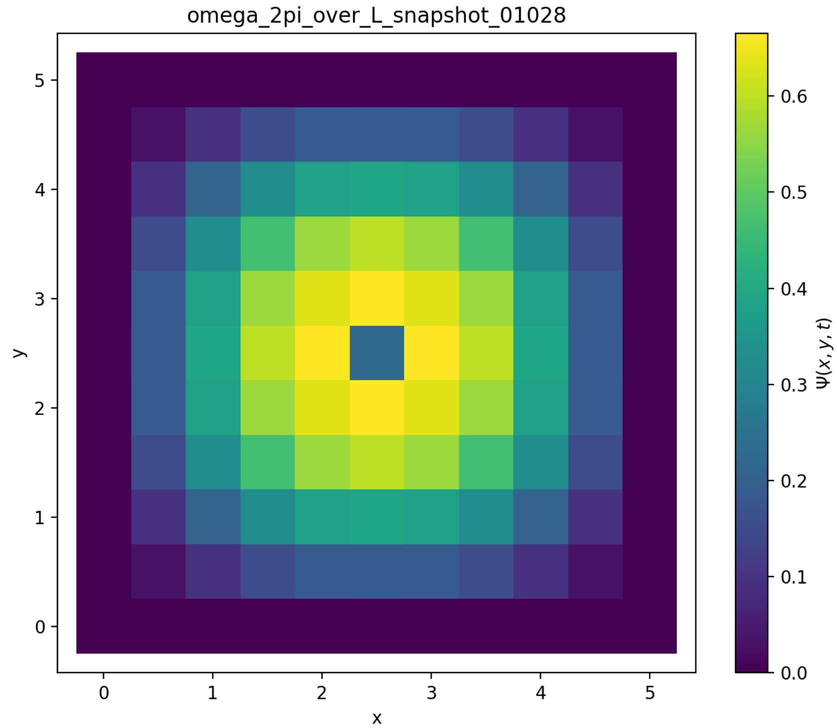

Figure 4: Heatmap of the membrane displacement for $\omega = 2\pi/L$.

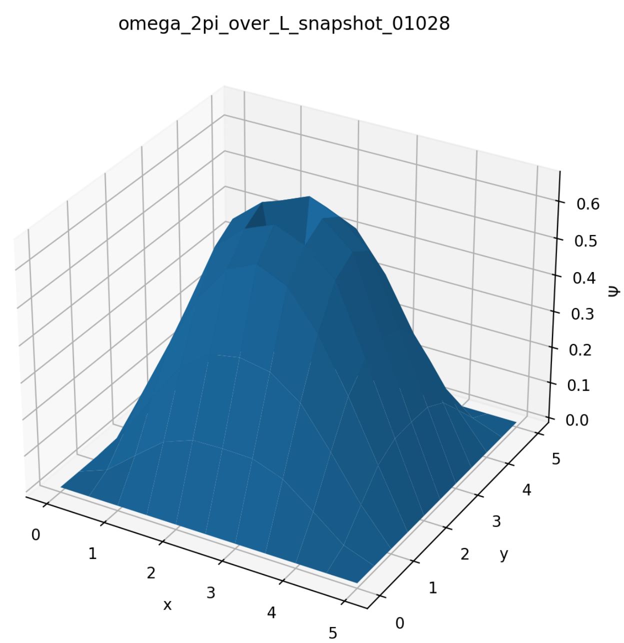

Figure 5: Three-dimensional surface plot of the membrane displacement for $\omega = 2\pi/L$.

The maximal oscillation amplitude as a function of time is shown in Fig. 6. Among all investigated frequencies, this case produced the strongest growth of oscillation amplitude.

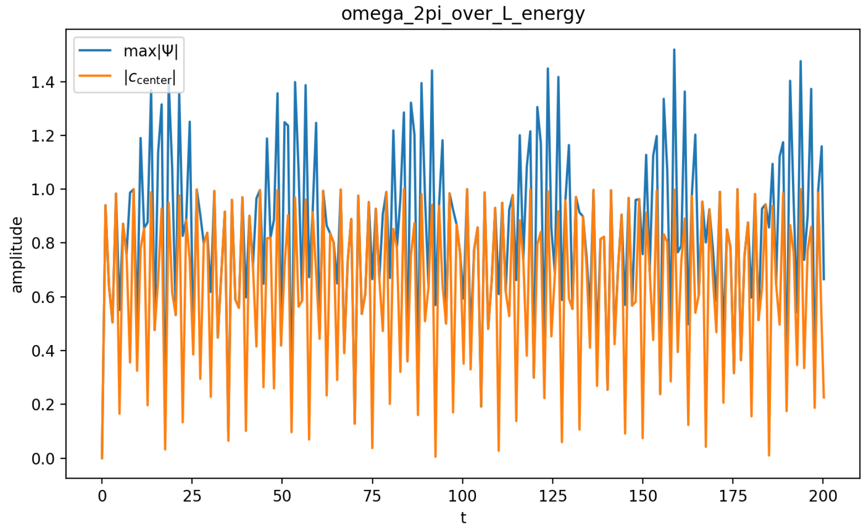

Figure 6: Maximum membrane amplitude as a function of time for $\omega = 2\pi/L$.

2.3 Oscillations for $\omega = \pi/(2L)$

Finally, the system was investigated for the lower driving frequency

$$\omega = \frac{\pi}{2L}.$$

In this case the membrane response is smoother and dominated by lower-frequency spatial modes. The amplitudes remain noticeably smaller than in the resonant regime.

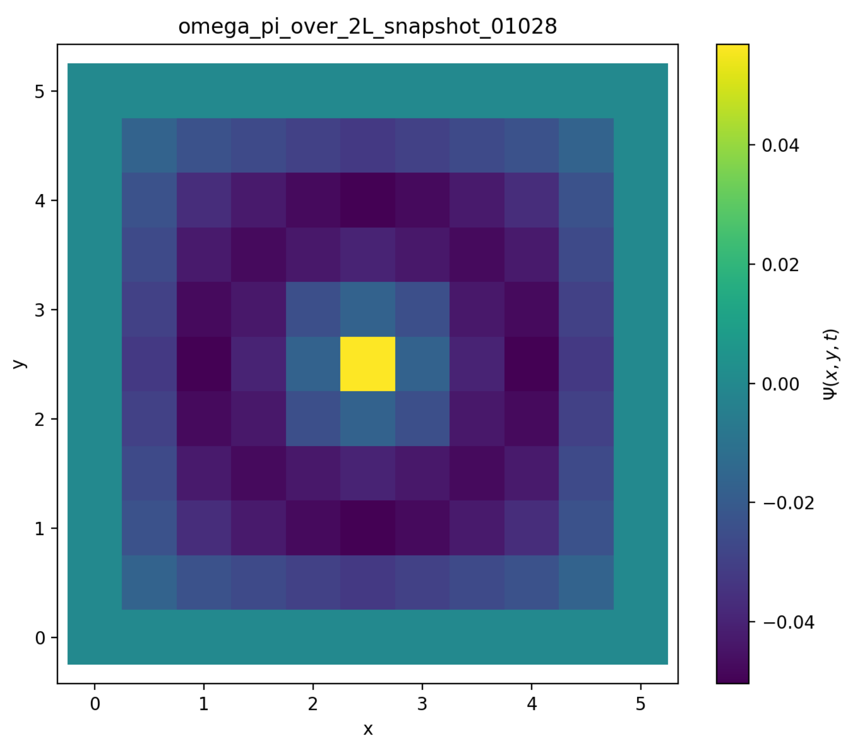

Figure 7: Heatmap of the membrane displacement for $\omega = \pi/(2L)$.

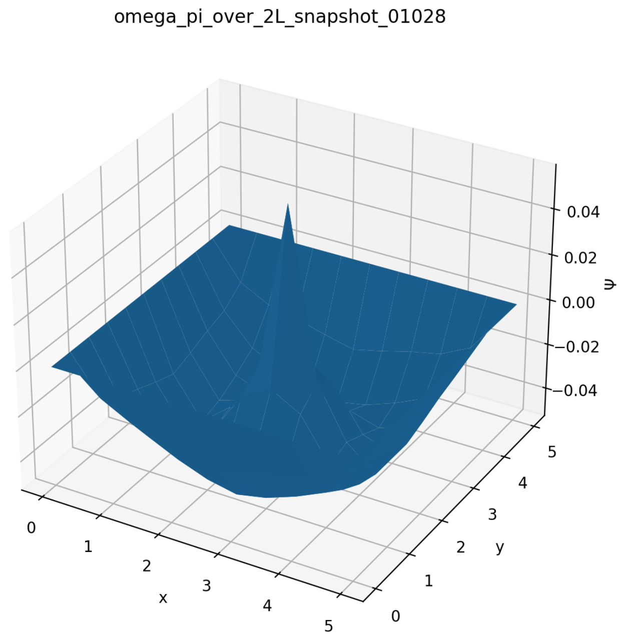

Figure 8: Three-dimensional surface plot of the membrane displacement for $\omega = \pi/(2L)$.

The time dependence of the maximal oscillation amplitude is shown in Fig. 9. The increase of oscillation amplitude is weaker than in the previous cases.

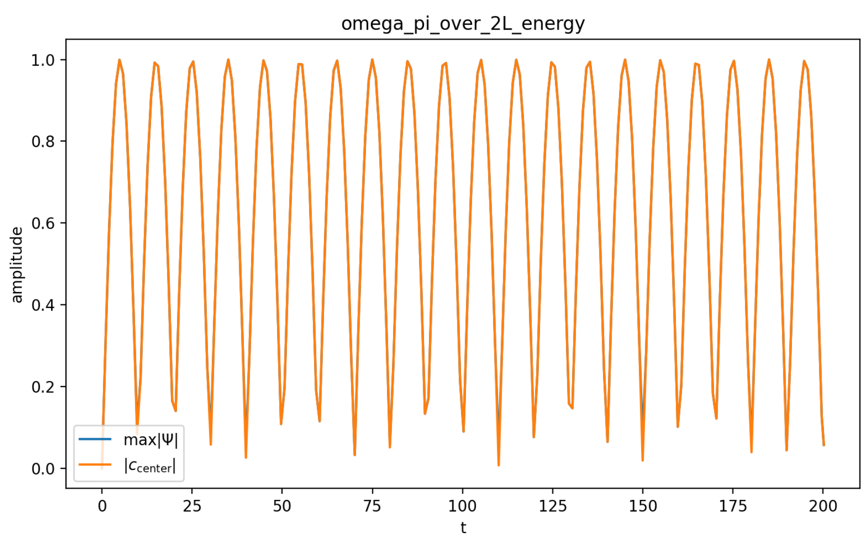

Figure 9: Maximum membrane amplitude as a function of time for $\omega = \pi/(2L)$.

2.4 Comparison of different driving frequencies

Figure 10 compares the maximal membrane amplitudes obtained for all investigated forcing frequencies. The strongest response was observed for $\omega = 2\pi/L$, indicating that this frequency is closest to one of the natural oscillation modes of the membrane.

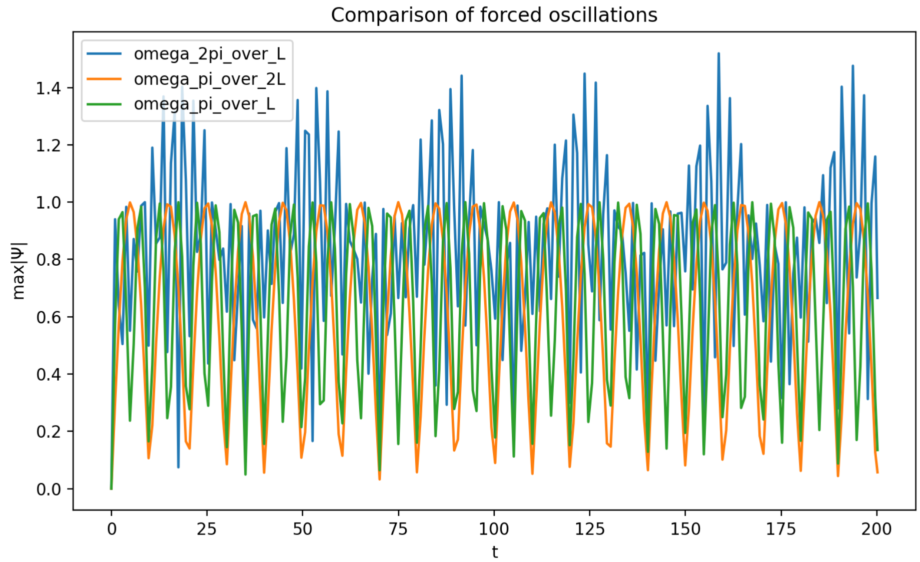

Figure 10: Comparison of maximal membrane amplitudes for different forcing frequencies.

The simulations demonstrated that the system does not reach a true stationary state. Since no damping mechanism was included in the model, energy is continuously supplied to the membrane by the external forcing. Consequently, the oscillation amplitudes continue to increase in time, especially for frequencies close to resonance.

3 Conclusions

In this project the classical wave equation for a two-dimensional square membrane was solved numerically using the finite element method combined with an explicit central-difference time integration scheme. The spatial discretization was performed with bilinear finite elements, which allowed the construction of the stiffness and overlap matrices required for the FEM formulation of the problem.

The stability condition for the explicit method was determined numerically from the largest eigenvalue of the generalized eigenproblem involving the stiffness and overlap matrices. A sufficiently small time step ensured stable long-time evolution of the membrane oscillations.

The simulations demonstrated the formation of standing-wave patterns resulting from reflections at the fixed boundaries of the membrane. Different forcing frequencies produced significantly different oscillation behaviors. The strongest resonance effects were observed for the driving frequency

$$\omega = \frac{2\pi}{L},$$

for which the membrane amplitudes increased most rapidly. Lower driving frequencies generated smoother oscillation patterns and weaker amplitude growth.

The calculations also showed that the system does not reach a stationary state. Since the model did not include any damping mechanism, energy was continuously supplied to the membrane by the periodic forcing applied at the central node. As a consequence, the oscillation amplitudes continued to increase in time, especially near resonant frequencies.

Overall, the finite element approach provided stable and physically meaningful solutions of the two-dimensional wave equation and allowed detailed investigation of forced oscillations and resonance phenomena in a square membrane.